

CS 490-002 - Computer Science Capstone

CRN: 24074

Spring 2026

3 Credit Hours

Instructor

Dr. Christopher Harris

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Email: christopher.harris@unco.edu

(Note: Please include "CS 490" in the subject line of all course-related emails to ensure a timely response.)

Office Hours

- Tuesdays & Thursdays: 10:00 – 11:00 AM (and typically immediately after class)
- By appointment (Zoom or phone available with advance notice; evenings usually work best)

Course Meetings

- Time: Tuesdays 2:00 – 3:15 PM
- Location: 2295 Ross Hall

Required Text

None; however, readings will be assigned to supplement lectures.

Prerequisites

- CS 301 (Algorithms and Data Structures) with a minimum grade of C
- Must be enrolled in one of the following Programs:
 - Computer Science-BS
 - Mathematics: Computer Sci-BS

Must be enrolled in one of the following Classifications:

- Junior
- Senior

Catalog Course Description

This course provides the opportunity to integrate the knowledge gained from courses taken in the Computer Science curriculum. An exploration of important topics in Computer Science will be discussed. In addition, students will work in teams to produce software programs and related artifacts by applying one or more established software frameworks and following a software development methodology.

Course Learning Objectives

Upon successfully completing this course, students will be able to:

1. Integrate multiple areas of computer science into a cohesive software system.
2. Apply an appropriate software development methodology at a small-team scale.
3. Design, implement, and test a medium-sized software application.
4. Produce professional technical documentation and design artifacts.
5. Collaborate effectively while demonstrating individual ownership.
6. Communicate technical work clearly to technical and non-technical audiences.
7. Analyze ethical and societal implications of computing solutions.
8. Reflect on professional readiness and career preparation.

Grade Determination

- Project Proposal & Planning – 5%
- Requirements & Architecture – 15%
- Weekly Professional Contribution – 20%
- Midterm Demo & Technical Review – 15%
- Final Software System – 25%
- Final Documentation & Reflection – 10%
- Final Presentation & Demo – 10%

Letter Grades

A: 90% and above
B: 80% – 89.99%
C: 70% – 79.99%
D: 60% – 69.99%
F: Below 60%

- Plus/minus grades may be used at the instructor's discretion.
- Grades may be curved based on overall distribution.
- Incomplete ("I") grades are given only in cases of documented severe hardship.

CS 490 Grading Rubric (Fair & Small-Cohort Appropriate)

1. Project Proposal & Planning (5%)

Criteria	Excellent	Satisfactory	Needs Improvement
Scope & Feasibility	Clear, realistic, well-scoped	Minor scope issues	Unrealistic or vague
Roles & Responsibilities	Clearly defined	Partially defined	Unclear ownership

2. Requirements & Architecture (15%)

Criteria	Excellent	Satisfactory	Needs Improvement
Requirements	Clear, testable, complete	Minor gaps	Missing or vague
Architecture Design	Appropriate, justified	Mostly sound	Poorly designed
Diagrams & Documentation	Clean, professional	Adequate	Incomplete

3. Weekly Professional Contribution (20%)

Assessed individually.

Evidence	Expectations
Git Contributions	Regular, meaningful commits
Task Ownership	Leads assigned components
Meetings	Prepared, engaged
Reliability	Meets deadlines consistently

Students who consistently fail to contribute may receive significantly reduced credit, regardless of the team's overall success.

4. Midterm Demo & Technical Review (15%)

Criteria	Excellent	Satisfactory	Needs Improvement
Functionality	Core workflows operational	Partial workflows	Minimal progress
Technical Understanding	Can explain design choices	Some uncertainty	Cannot explain
Code Quality	Organized, readable	Adequate	Disorganized

5. Final Software System (25%)

Criteria	Excellent	Satisfactory	Needs Improvement
Feature Completion	All core features working	Most features	Major gaps
Code Quality	Clean, modular	Acceptable	Poor structure
Testing	Meaningful tests	Minimal tests	Little or no testing
Robustness	Handles errors	Some edge cases	Fragile

6. Final Documentation & Reflection (10%)

Criteria	Excellent	Satisfactory	Needs Improvement
Technical Report	Clear, concise, complete	Adequate	Weak or missing
Reflection	Insightful, specific	General	Superficial

7. Final Presentation & Demo (10%)

Criteria	Excellent	Satisfactory	Needs Improvement
Clarity	Clear narrative	Mostly clear	Confusing
Demo Quality	Polished, stable	Minor issues	Unreliable
Team Balance	All members contribute	Uneven	One-person demo

Student Success

I find that most students who wait until the last minute to submit homework and programming assignments often don't do as well as those who start on assignments early. Try to set a goal of starting homework and programming assignments as soon as they are assigned, look over what is being asked, and think about it before beginning. This will help you come up with questions to ask, and you will perform better on the assignment.

It is important to me that you have every opportunity to excel in this class, so please come see me, email me, or meet me on Zoom if you run into any difficulties. When you come to see me, please explain what you have tried and the results you obtained, and be prepared to demonstrate them during our meeting. The more that you have tried in advance and the more specific your questions are, the better I can help you.

Again, please do not email me when you miss a class and ask what you missed – you can obtain this information from another student, and important deadlines appear in course

announcements and in Canvas. However, other helpful information and reminders are conveyed in class meetings.

This course requires you to work hard, and as a 400-level class, it demands considerable effort outside the classroom; success requires a significant time commitment.

Attendance

A lot of information needs to be conveyed, so missing class negatively affects the learning process.

Here is the university policy on attendance:

Students are expected to attend class regularly. Each instructor determines the relationship between class attendance, the objectives of the class, and students' grades. Instructors are responsible for articulating their attendance policies and their effect on grades to students. Students are responsible for knowing the attendance policy of each course. Only the instructor can approve students' absences. Students are responsible for requesting such approval. In an effort to create inclusive learning environments, instructors should not require doctors' notes to determine whether or not to excuse an absence.

Quizzes may be given on material covered in class if attendance is low or if the instructor believes the students are not prepared.

Please wait up to 15 minutes if I am late to class.

Making Up Classwork

Attending class is important to do well in this class. During the first week of class, exchange contact information with your classmates. Although I appreciate you telling me when you will not be in class, please don't ask what you missed. It is better to obtain this from other classmates.

Instructor Absences

I have an active research program in Computer Science. This requires me to be absent from the classroom and campus on occasion, and classroom sessions will not be held, but the learning process will continue. These planned absences will be communicated in advance through Canvas announcements and during our class meetings. On those dates, online lectures (if any) and assignments may be done, but there is no required physical classroom attendance.

Readings

Readings may include one or more chapters each week. It is expected that the readings given each week in the course schedule will be read *before* the class meets for that topic. Make sure that you understand the reading each week to keep up and to achieve good academic performance – if you are unfamiliar with a concept or topic, email me, stop by office hours, or

bring it up in class. Some readings will refer to materials and concepts not mentioned in the course notes; it is the student's responsibility to become familiar with the referenced material.

Late and Not-Completed Assignments

When you get on the job or go to graduate school, meeting deadlines will matter a lot. Contracts and jobs are lost if a deadline is missed, and the hard work of you and your colleagues can go unrewarded. All submissions are time-stamped and considered late after the indicated deadline.

Assignments are to be submitted through the dropbox. Most assignments are due at 11:59 PM on the date specified (send the assignment by email to the instructor at christopher.harris@unco.edu if with "CS 490" in the title only if the Canvas dropbox is closed). Note: You will still need to submit it through Canvas, but the email with the assignment establishes a timestamp.

If the date given in the syllabus or assignment sheet differs from the date given by your instructor in class, by announcement, or on Canvas, the information given in class/by announcement/Canvas will be taken as the official deadline.

It is best to submit early to ensure your assignment successfully uploads to Canvas and is properly attached to the correct dropbox. Ensure (1) the correct attachment is made, (2) it is put in the right Canvas dropbox. Your most recent submission will be used if you submit more than one version of your assignment. If the assignment is not included in the dropbox (or in the email to the instructor), it has not been received, and late penalties will still apply. In many cases, providing a link to an external repository (e.g., GitHub) might be fine; in others, a Dropbox submission of a PDF or Word document is the only allowed file type.

Problems with the dropbox are not valid excuses for inability to submit an assignment – you can email me your submission (put 'CS 490' in the subject line for all emails regarding this course; if not, a response to your email is likely to be delayed). This establishes a time stamp for your submission. You will still need to upload the assignment to the Canvas dropbox as soon as possible, as that is where it will be graded.

I expect you to ensure your UNCO BEARS email works and that you check your email regularly as announcements are sent to you in this manner. Expect to get announcements from me throughout the semester. In these announcements, some due dates and assignments may be changed from what is in the syllabus, so it is important to check them regularly.

Late Penalties

Late assignments are penalized as follows:

Hours late	Consequence
< 24	10% penalty (rounded to the nearest 0.5 point)
24 – 48	20% penalty (rounded to the nearest 0.5 point)
> 48	Not accepted, factored into the overall grade as a 0

You may assume that all submissions may be submitted late (with the applicable penalty) unless stated otherwise.

Keep in mind that it may make more sense to turn in a “better” submission a day late (and earn up to 90% of the points available) than submitting a half-completed assignment before the original deadline (and only earn 50%). Use your own discretion as to which makes more sense in your case.

Groupwork

This course will involve groupwork. In some cases, you will get to choose your groupmates and in other cases they will be assigned. Again, this reflects what happens on the job; your ability to work with all kinds of people is important to employers and to your colleagues (and hopefully to you as well).

If the group dynamics involve conflict that cannot be resolved within your group, please come and talk with me. Please be prepared to discuss what you tried to do to resolve the situation. Feel free to bring a classmate or friend with you if this makes you feel more comfortable when you come talk with me. In some cases, it may be a simple misunderstanding; in others, it may need more significant adjustments.

Do understand that this course also assesses your communication skills, both in written and oral. Therefore, on exams, homework assignments, and projects, the use of proper written grammar is expected.

Classroom Behavior

Don't be rude. Please be considerate of your fellow classmates and do not act in a disruptive manner. Turn off or silence your cell phones before coming to class and keep them put away, and arrive on time. When I am lecturing, side conversations between students are noticeable and will likely be called out by me, affecting the class participation component of your grade. During labs and other exercises, conversations among students are fine as long as they do not distract others. When working in groups, try to keep the conversation focused on the task at hand. If you need to leave the room for any reason (like using the restroom), please do so as quietly as possible.

Academic Integrity

It is critical that each student does his or her own work and properly acknowledges the work of others.

Here is the university policy on academic integrity:

Students are expected to practice academic honesty in every aspect of this course. Students who engage in academic misconduct are subject to grading consequences with regard to this course and/or university disciplinary procedures through the Student Conduct and Accountability. More information about the academic misconduct process can be found in UNC's Student Code of Conduct (BEAR Code).

I expect students to be honest and not cheat on their assignments and exams. The exams must be completed without giving or accepting assistance from other students. In programming assignments, any source code copied from another source must be credited as such.

When re-using materials, citing examples, or using quotes, it should always be clear to the instructor when relying on the work or results of others. If in doubt, be explicit in your assignments and reports. When accessing www resources such as downloadable software, technical reports, papers, online tutorials, etc., do not forget to read their accompanying legal notices and comply with their provisions. If you have questions or need further clarification, please ask me!

Open-source software used must maintain all required headers and other information as specified by the open-source license. I expect you to be familiar with the University's policies on student conduct, academic dishonesty, and related matters. UNC's policies and recommendations for academic misconduct will be followed.

For additional information, please see the Dean of Students' website, the Student Handbook link, and the current catalog. If I have evidence of cheating or plagiarism, I will vigorously follow the procedures outlined in the university policy. Plagiarism and exam cheating are particularly serious offenses and grounds for failure in the course. As stated in university policy, instances of academic integrity violations will be reported.

Responsible AI Policy

This course permits the use of artificial intelligence (AI) tools (e.g., code assistants, text-generation tools, and debugging aids) in a responsible and transparent manner. AI tools are intended to support learning, not replace it.

Permitted Uses

- Brainstorming design ideas or alternative approaches.
- Debugging assistance and explanation of compiler/runtime errors.
- Generating boilerplate code that is fully reviewed, modified, and understood by the student.
- Improving clarity and grammar in documentation written by the student.

Prohibited Uses

- Submitting AI-generated code or documentation without understanding or review.
- Using AI tools to generate complete assignments or project components wholesale.

- Using AI to bypass collaboration, learning, or peer contribution expectations.
- Presenting AI-generated work as original student reasoning.

Disclosure Requirement

Students must disclose meaningful AI usage in a short appendix or comment within their submission, describing how the tool was used and what modifications were made. Failure to disclose may be treated as an academic integrity violation.

Accountability

Students are fully responsible for the correctness, security, ethics, and originality of all submitted work, including any portions assisted by AI tools. AI-generated content does not excuse errors, plagiarism, or policy violations.

Instructor Use

The instructor may use Generative AI for feedback, examples, and assessment design.

Disabilities

The following is the current UNC policy on disability support:

It is the policy and practice of the University of Northern Colorado to create inclusive learning environments. If there are aspects of the instruction or design of this course that present barriers to students' inclusion or to accurate assessments of students' achievement (e.g. inaccessible web content, use of videos without captions), students should communicate about these aspects with their instructor(s). Additionally, if you have a temporary health condition or a permanent disability that requires accommodations, contact the Disability Resource Center (DRC) as soon as possible. DRC facilitates the interactive process that establishes reasonable accommodations.

Office: (970) 351-2289, Michener Library L-80. Students can learn more here: www.unco.edu/disability-resource-center

Modifications to the Syllabus

I reserve the right to modify the syllabus and assessment criteria in response to student needs, developments, and unforeseen events.

Tentative 16-Week Schedule

Formal class meetings occur on Tuesdays only. All other time is reserved for student-led meetings, development, and research.

Week	Date	Tuesday Focus	Primary Activities / Deliverables	Grading Component(s)	ABET Outcome(s)
1	Jan 13	Course Orientation	Course overview, expectations, tooling, team roles, project brainstorming	Participation, Weekly Contribution	ABET 5, 7
2	Jan 20	Project Selection & Proposal	Finalize project scope, roles, feasibility analysis; submit proposal	Project Proposal & Planning	ABET 2, 5
3	Jan 27	Requirements Engineering	User stories, functional & non-functional requirements; Software Requirements Specification (SRS) draft	Requirements & Architecture	ABET 1, 2
4	Feb 3	Architecture & Design	System architecture, technology stack, UML diagrams	Requirements & Architecture	ABET 2, 6
5	Feb 10	Sprint 1 Planning	Sprint planning, task assignment, initial implementation	Weekly Professional Contribution	ABET 5
6	Feb 17	Sprint 1 Review	Code walkthroughs, integration check, scope adjustment	Weekly Professional Contribution	ABET 1, 6
7	Feb 24	Sprint 2 Planning	Testing strategy, deployment considerations, feature expansion	Weekly Professional Contribution	ABET 3, 5
8	Mar 3	Midterm Demo	Live demo of core workflows; technical review and feedback	Midterm Demo & Technical Review	ABET 1, 6
9	Mar 10	Refinement & Testing	Address feedback, testing, and documentation updates	Weekly Professional Contribution	ABET 3
10	Mar 17	Spring Break	No class meeting	—	—
11	Mar 24	Sprint 3 Development	Advanced features, robustness, optional stretch goals	Weekly Professional Contribution	ABET 1, 3
12	Mar 31	Ethics & Professional Practice	Ethics, security, and responsible AI discussion	Weekly Contribution, Documentation	ABET 4
13	Apr 7	Final Integration & Testing	Full system integration, bug fixing, and regression testing	Final Software System	ABET 1, 3
14	Apr 14	Final Preparation	Presentation rehearsal, documentation finalization	Final Documentation & Reflection	ABET 6
15	Apr 21	Final Presentation & Demo	Live demo, presentation, peer evaluations	Final Presentation & Demo; Final Software System	ABET 5, 6
16	Apr 28	Reflection & Wrap-Up	Individual reflection, lessons learned, course wrap-up	Final Documentation & Reflection	ABET 7

ABET Outcome Key:

1 – Analyze and solve complex computing problems

- 2 – Apply computing design and development principles
- 3 – Communicate effectively in professional contexts
- 4 – Recognize ethical and social responsibilities
- 5 – Function effectively on a team
- 6 – Develop and conduct evaluation/testing
- 7 – Engage in professional development and reflection

Course Withdrawal Deadline: May 1, 2026

Last Updated: 1/7/26